

Process Documentation

GITHUB REPO - <https://github.com/Mdot5596/MA-Dissertation>

Documentation Method

The development process for this project was documented through GitHub Version Control, prototype documentation posters and the use of a Trello board (links in the viva power point. GitHub was used throughout development to store the project files and maintain a record of changes made during the development of the prototypes.

The prototype posters were also used as a form of process documentation. Each poster documents the purpose, design decisions, techniques and findings associated with each prototype, providing a visual record of how the project developed and how the different environmental guidance techniques were implemented and evaluated.

I also discussed the approach of process documentation with my supervisor during the supervisor meetings. Based on this discussion, I used the evidence already produced throughout development, including GitHub Commits, prototype screenshots, posters and testing results to demonstrate the progression of the project rather than creating a separate development blog.

The following timeline provides a chronological overview of the project's development, with screenshots and supporting evidence showing how the artefact progressed from the initial research and concept throughout to the final prototypes and testing.

Development Timeline

1. Initial Research & Project Concept

The project began with research into player navigation, environmental guidance and wayfinding in games. This research was used to identify potential environmental techniques that could influence player movement without relying on explicit guidance systems such as objective markers, arrows, or minimaps.

This research led to the development of the research question

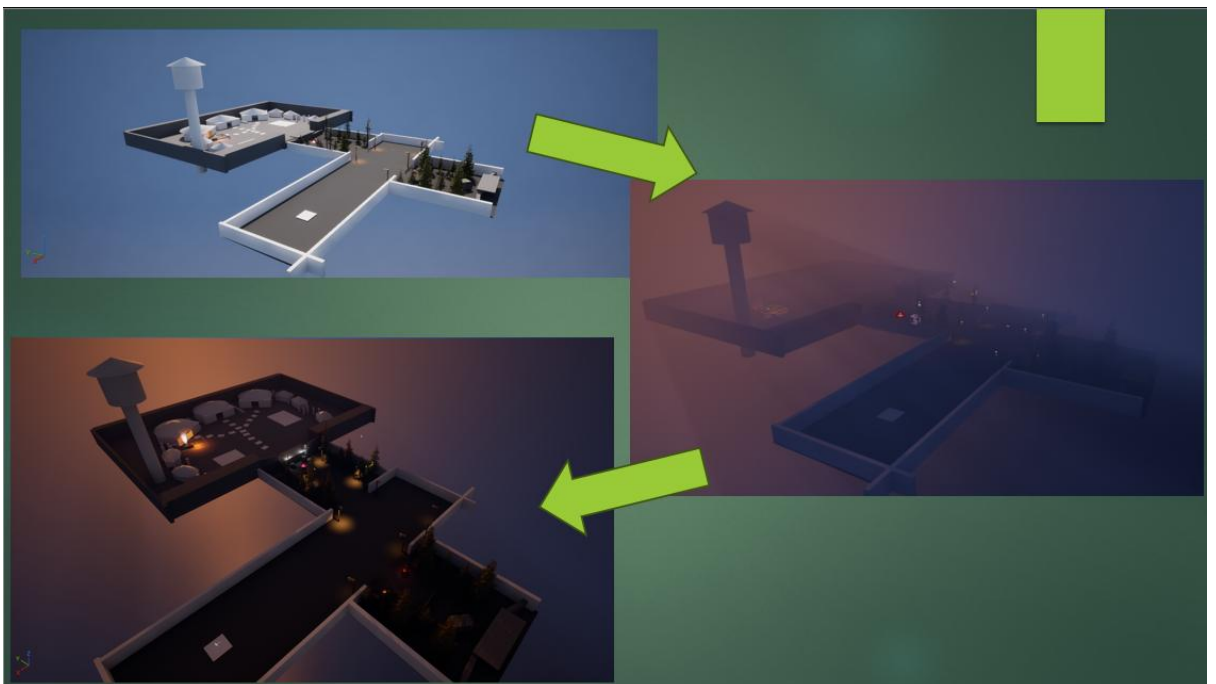
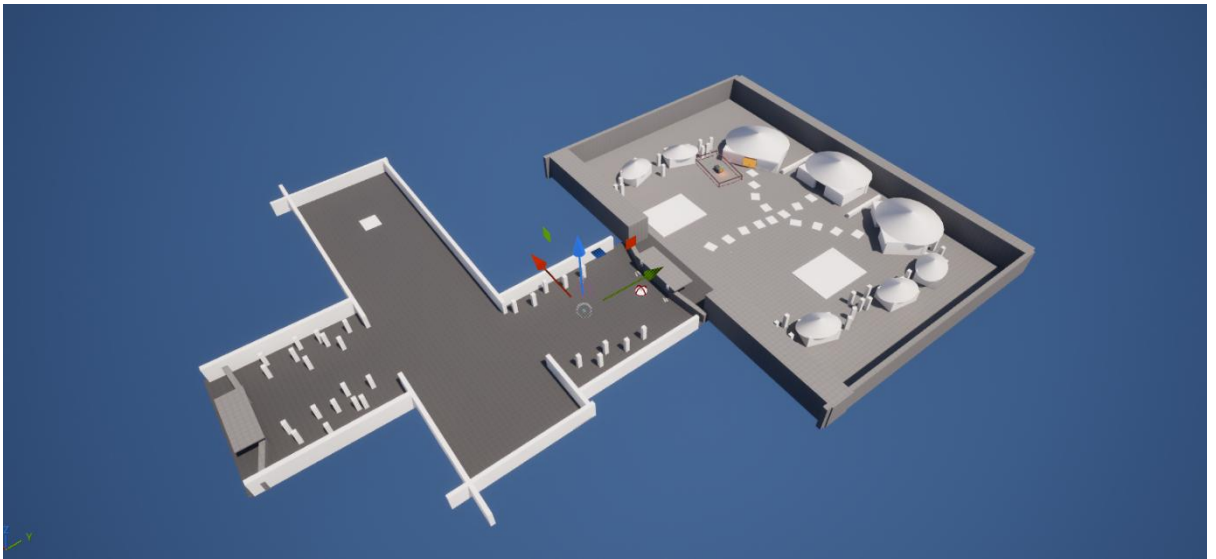
See 04_Research to access the Guidance Framework Document created at this time

01_Final_Artifact	25/08/2026 16:04	File folder
02_Project_Documentation	07/09/2026 22:22	File folder
03_Videos	07/09/2026 22:01	File folder
04_Research	07/09/2026 20:27	File folder
05_Testing	07/09/2026 20:21	File folder

2. Prototype 1 – Lighting Guidance

The environment was designed around contrast between darker areas and illuminated areas, using light to create visual hierarchy and draw attention towards the intended route.

During development, different lighting placements and environmental elements were tested before creating the final prototype.



Prototype 1: Lighting guidance

Using Light to Guide Player Navigation

Using light, contrast and illuminated points of interest to influence player movement without explicit UI guidance

1) Dark Environment
The environment is intentionally presented at night, reducing visibility and making illuminated areas more visually prominent



2) Light Contrast
Brighter areas contrast the surrounding darkness, naturally attracting the player's attention and suggesting areas worth exploring



3) Directional Lighting
Lighting is positioned along the intended route to subtly encourage forward movement without directly indicating an objective



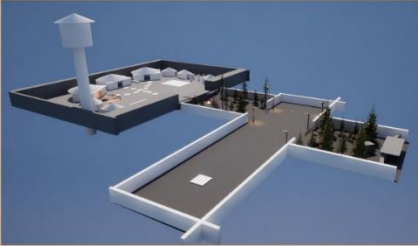
4) Focal Point/Reveal
A stronger light source is introduced near the destination, creating a visual focal point that can be recognised from further away



5) Environmental Feedback
The generator provides visual feedback through its orange light, smoke, and sparks



Before - Low visibility makes the environment difficult to read and provides few obvious points of interest



After - increased lighting contrast creates visual hierarchy and draws attention towards the intended route and destination



Key Findings

Lighting effectively attracted player attention and influenced initial navigation. Players consistently gravitated towards illuminated areas, with all recorded testers reaching the objective. This suggests that strong lighting can act as an effective directional cue without explicitly telling the player where to go.

Key Guidance Techniques

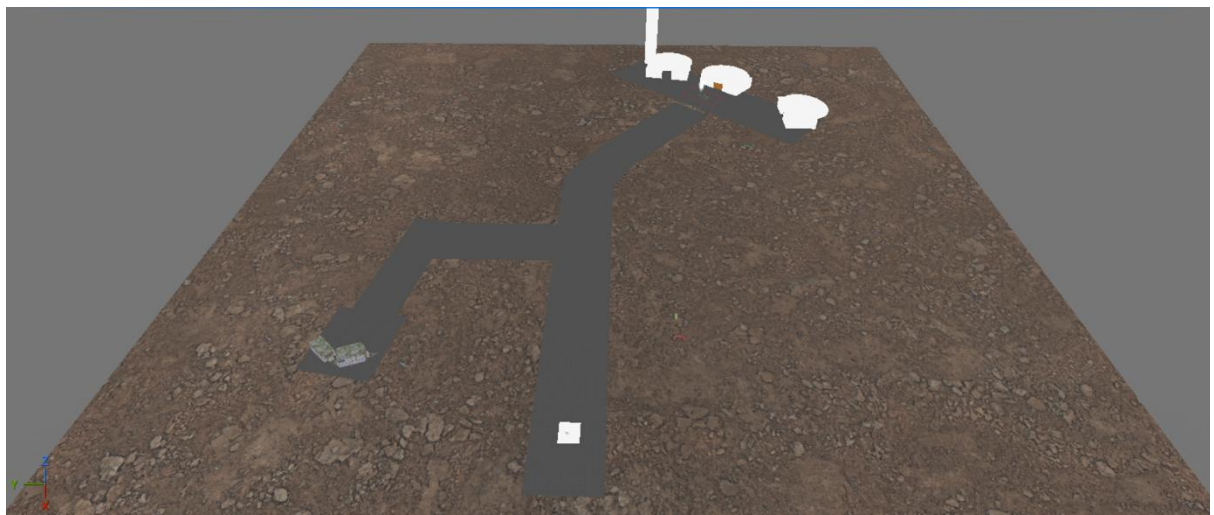
- Light/Dark Contrast
- Directional Lighting
- Focal Points
- Environmental Feedback
- Warm Lighting
- Limited Visibility
- Valves
- Forks

3. Prototype 2 – Architectural Guidance

The second prototype investigated how the physical layout of an environment could influence player route selection.

The prototype explored techniques including path width contrast, compression and release, framing, environmental funnels and reveals.

The design was developed to reduce reliance on lighting or obvious landmarks and instead test whether the structure and layout of the environment could communicate a preferred route.





Prototype 2: Architectural guidance

Using Layout to Guide Player Navigation

The environments architecture and layout influence player movement without the use of lighting emphasis, landmarks or UI

1) Wide, Open Start

The player starts in an open area that naturally encourages forward movement



2) Fork: Path Width Contrast

The correct path is wider, cleaner and more comfortable. The wrong path is tighter and feels more enclosed



3) Compression

The path narrows between rocks and trees, creating compression and building anticipation before the reveal



4) Reveal

The player moves through a restricted section of the environment before the path opens up, revealing the camp area. This creates anticipation and uses the structure of the environment to control when the player sees the destination



5) Framing & Flow

Trees and rocks frame the playable space and subtly encourage the player to follow the intended route. Curved paths and environmental boundaries maintain forward momentum while limiting unnecessary exploration





Key Guidance Techniques

- Path Width Contrast
- Compression and release
- Curved Paths
- Framing
- Occlusion (limited sightlines)
- Natural Funnels (Trees/Rocks)
- Environmental Reveals

Wrong Path (Dead End)

The wrong path is narrow, enclosed and leads to a blocked area with no reward



Key Findings

Architectural design provided the strongest and fastest route guidance of the three prototypes. Players were influenced by differences in path size, spatial layout and environmental framing, with all recorded testers reaching the objective. The results suggest that the structure of the environment itself can strongly influence player route selection.

4. Prototype 3 – Initial Landmark Guidance

The third prototype investigated whether distinctive environmental landmarks could help players orient themselves and navigate through an environment. The initial design included landmarks such as the water tower and windmills.

However, the first round of testing revealed an important problem with the prototype.

Players were able to use roads and paths as navigational cues, meaning it was difficult to determine whether they were actually responding to the landmarks or simply following the architectural layout of the environment.

This was an important point in the development process because it demonstrated that the guidance technique could not be effectively tested without considering the other environmental information available to the player.



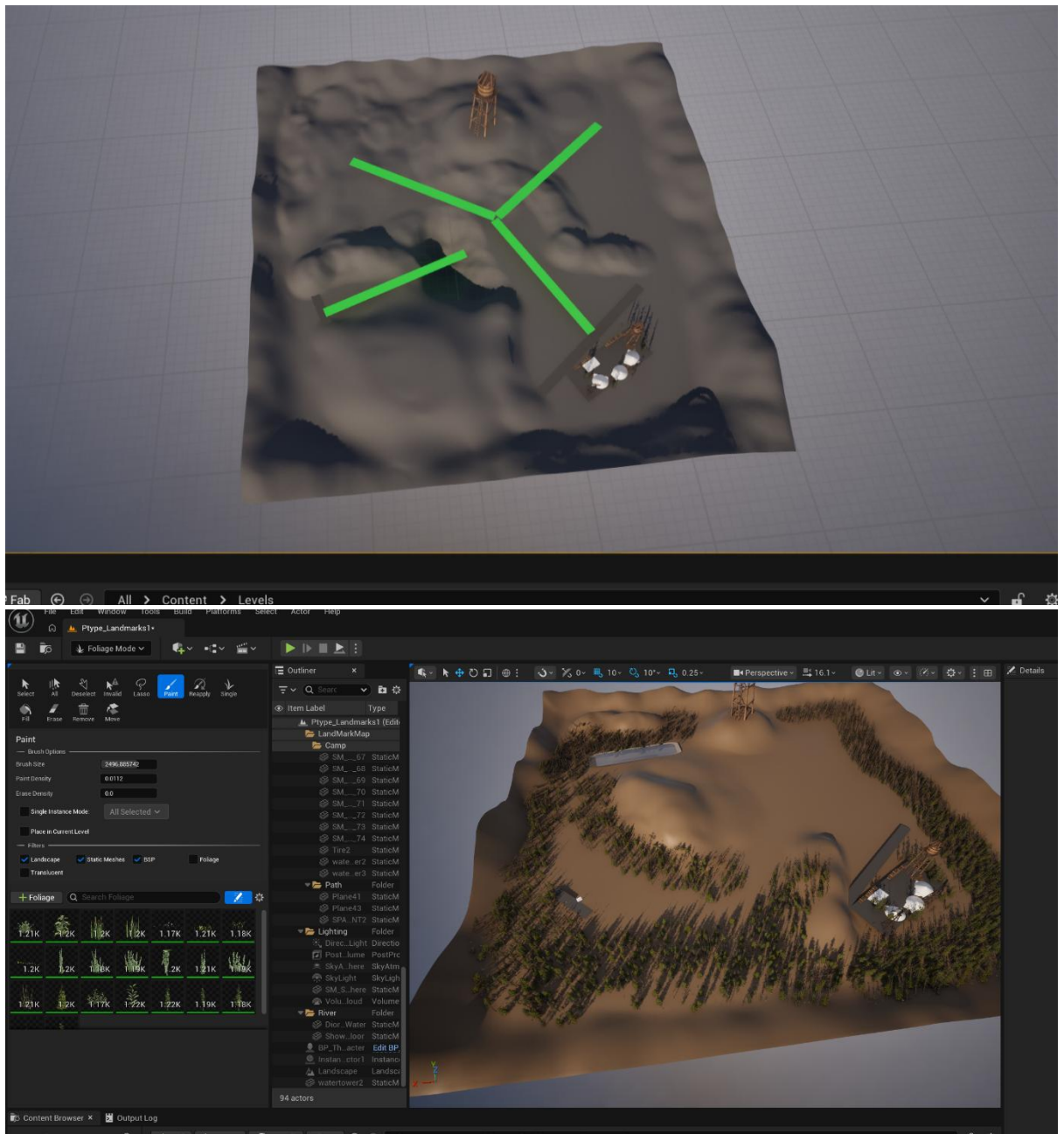
5. Prototype 3 – Iteration & Redesign

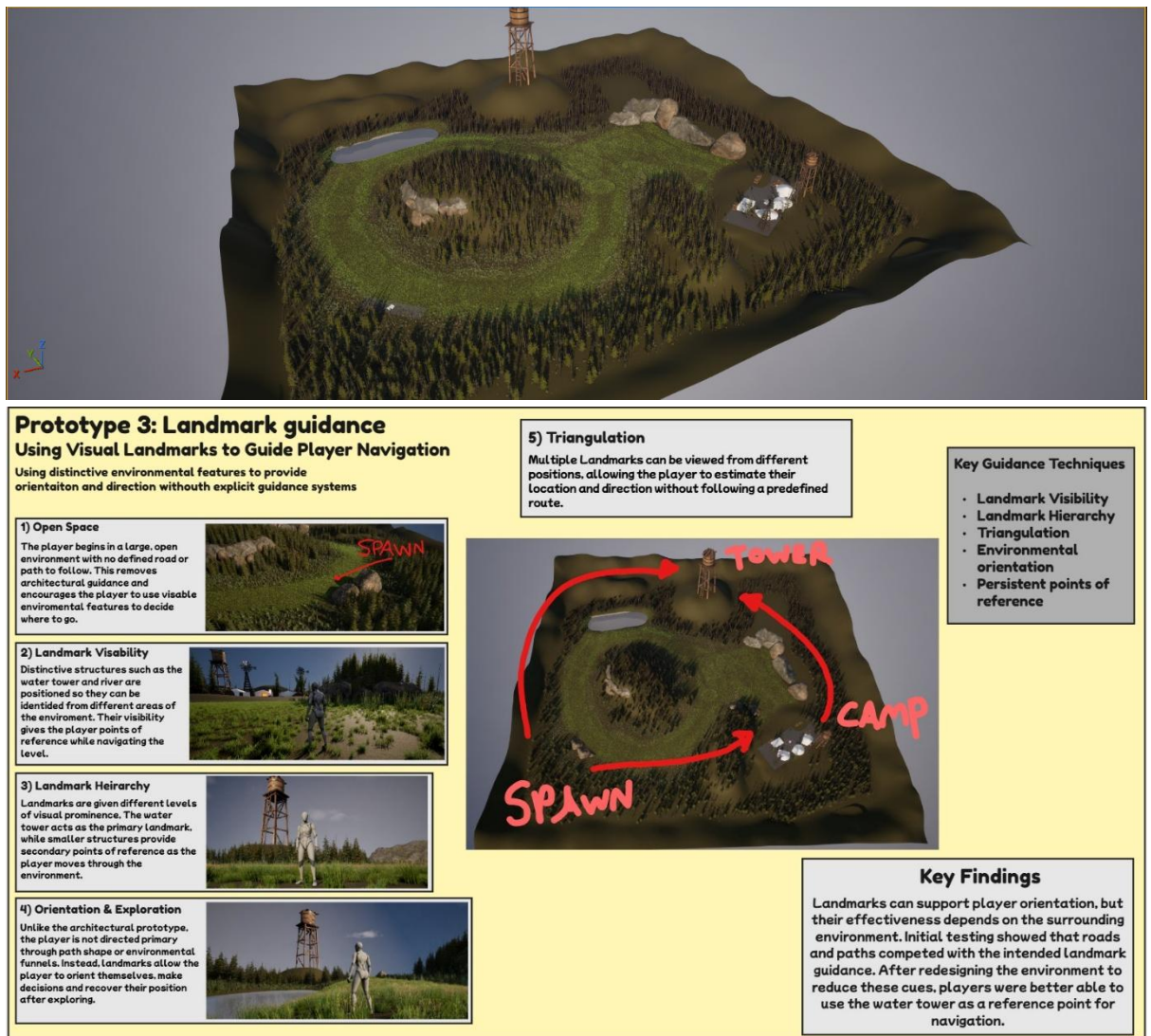
Following the initial testing, Prototype 3 was redesigned to reduce the competing architectural guidance.

The redesigned environment used a more open natural landscape with fewer obvious roads and paths. The water tower was made the primary landmark, while other landmarks were used to provide additional environmental reference points.

The purpose of this change was to make it more likely that players would rely on the landmark itself when making navigation decisions.

This iteration became one of the most significant developments in the project because the testing directly influenced the design of the artefact.





6. Player testing

Player testing was conducted across the prototypes to investigate how players responded to the different environmental guidance techniques.

Testing recorded factors including

- Objective completion
- Completion time where available
- Navigation confidence
- Perceived difficulty
- Player feedback
- Which environmental cues players appeared to use

The results were then used to compare the different techniques and identify areas where the prototypes required further development.

Testing & Findings

Evaluating Environmental Guidance Techniques

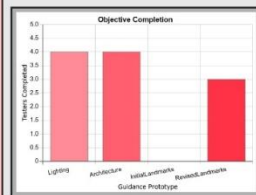
Testing Method

Participants were asked to navigate each prototype from a fixed starting point to a designated objective without being told which environmental guidance technique was being investigated

Measures

- Objective completion
- Completion Time
- Navigation Confidence
- Perceived difficulty
- Guidance technique identified
- Player observations and feedback

Results

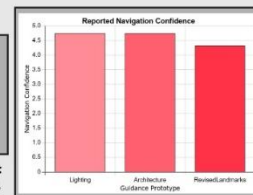


Both Lighting and Architecture produced more successful objective completion for all recorded participants. Initial Landmark testing revealed significant difficulty locating the objective, leading to a redesign of the prototype.

Average Recorded Completion Time

Prototype	Average
Lighting	52.7s
Architecture	28.0s
Landmarks – initial	125s
Landmarks – revised	96.3s

Architecture produced the fastest average recorded completion time, while the initial landmark prototype resulted in a longer navigation and difficulty locating the objective.



All three techniques generated relatively high reported confidence, although landmark navigation produced slightly lower confidence than lighting and architecture.

Key Findings

1 - Lighting attracted attention

Players consistently gravitated towards illuminated areas, suggesting that lighting can influence initial player direction without explicit instructions or any other guidance methods. Lighting can therefore be used independently or combined with other environmental techniques.

2 - Architecture provided efficient navigation

The architectural prototype produced the fastest average recorded completion time, with path width and environmental structure providing clear directional cues.

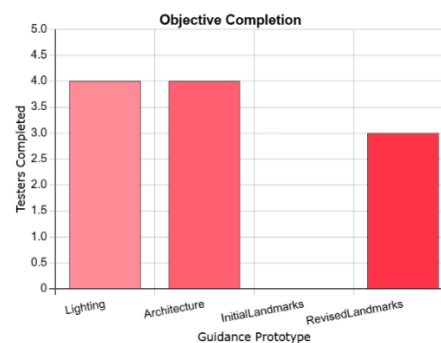
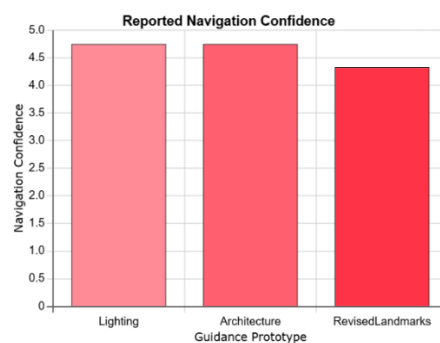
3 - Landmarks supported orientation

The initial Landmark prototype showed the difficulty of separating landmark guidance from architectural guidance. After redesigning the environment to reduce path-based cues, testers were able to use the water tower as a navigational reference. This was the most challenging guidance technique to isolate.

Overall Finding

The testing suggests that environmental design can influence player navigation without explicit UI guidance. Lighting successfully attracted player attention, while architectural design provided clear and efficient route guidance. Landmarks supported orientation, but their effectiveness depended more heavily on the surrounding environment and competing navigational cues.

Overall, the findings suggest that no single guidance technique is universally more effective. Instead, environmental guidance is influenced by how different cues are designed and interpreted within the wider level.



All Testing Docs can be found here:

Name	Date modified	Type	Size
01_Final_Artifact	25/08/2026 16:04	File folder	
02_Project_Documentation	07/09/2026 22:22	File folder	
03_Videos	07/09/2026 22:01	File folder	
04_Research	07/09/2026 20:27	File folder	
05_Testing	07/09/2026 20:21	File folder	